Output:

Enter how many numbers you want: 10

Fibonacci Sequence:

0 1 1 2 3 5 8 13 21 24

Date: 17/09/25

Task: 3 Implement Python generators & decorator.

AIM: Write a python program to implement python generator and decorators.

(3.1) :- Fibonacci Sequence Generator:

Algorithm: Write a python program to:-

- Create a generator function that calculates the execution time of any function.
- Apply the decorator to a function that sorts a list of random numbers.

Program:

```
# Fibonacci Sequence Generator using Generator
def fibonacci_generator (n):
    a,b = 0,1
    count = 0
    while count <n:
      a, b = b, a+b
      count +=1
num = int (input ("Enter how many numbers you want: "))
print ("Fibonacci Sequence:")
for value in fibonacci_generator (num):
    print (value, end = " ")
```

Result:

Output:

Sum is : 499999950000000

Execution Time : 0.2344834804534912 second.

Date: 17/09/25

Task (8.2) - Function execution Time Decorator

write a python program to :

- Implement a decorator that calculates the execution time of any function.

- Apply the decorator a function that sum a list of random numbers,

Program:

```
import time
def execution_time (func):
    def wrapper ():
        start = time.time()
        func ()
        end = time.time()
        #print execution time
        print ("Execution Time :", end - start, "second")
    return wrapper
@ execution_time
def demo_function ():
    total = 0
    for i in range (1, 10000000) :
        total += i
    print ("Sum is :", total)

demo_function()
```

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Result: The program successfully calculates the sum of numbers from 1 to 1000.00 and display the execution time using decorator function.